Fusion Reactor Neutronic Transport and Tritium Breeding in Layered Blanket Architectures: A Machine Learning-Assisted Approach

MPhys Research Project

A continuation of Semester 1 research on Fusion Breeder Blankets

Vadan Khan (10823198)
In collaboration with Rhodri Peters

4th Year MPhys, Department of Physics and Astronomy, University of Manchester

April 29, 2026

Abstract

The realisation of commercial nuclear fusion via the deuterium-tritium (D-T) fuel cycle is contingent upon achieving tritium production in breeder modules. Building upon our first homogeneous approximated studies, this study evaluates the neutronic performance of a heterogeneous layered breeder blanket architecture to provide a more physically realistic assessment, using $Li_{17}Pb_{83}$ breeder paired with pressurised water coolant. To navigate the computationally intractable 8-dimensional design space of layer thicknesses, a bespoke Single-Objective Bayesian Optimisation pipeline employing Gaussian Process Regression (SOBO-GPR) was developed, integrating parametric Computer-Aided Design (CAD) generation (Paramak) with stochastic neutron transport (OpenMC).

Results indicate that the transition to a layered architecture degrades the Tritium Breeding Ratio (TBR) due to the introduction of parasitic structural and coolant sinks. We sought to optimise individual layer thicknesses. The bespoke SOBO-GPR pipeline proved highly effective in navigating this complex space, identifying that optimal TBR demands a front-loaded, "binding configuration" of maximum breeder and minimum coolant thicknesses. Additionally, surrogate models and spatial reaction maps clearly reveal lateral considerations, such as the negligible impact of the outermost blanket layers on overall tritium production.

While this study establishes a robust breeding optimum, deeper neutronic analysis highlights the scope for more holistic study. The successful implementation of this modular machine learning pipeline provides a foundation for further research following this approach, where different materials could be easily configured. Integrating Multi-Objective Optimisation (MOO) and scaling up parameter spaces alongside the relaxation of geometric approximations could assist in the holistic balancing of TBR, dynamic heat extraction, and material longevity required for next-generation fusion reactors.

CONTENTS

1. INTRODUCTION

Commercial nuclear fusion is predominantly pursued via the deuterium-tritium (D-T) fuel cycle due to its relatively large fusion cross-section at achievable plasma temperatures compared to alternatives like D-D or D-^{3}He (Heckrotte and Hiskes, 1971; Paris and Chadwick, 2023). This reaction fuses deuterium and tritium to yield an alpha particle and a highly energetic fast neutron.

$$^2\mathrm{H} + {}^3\mathrm{H} \longrightarrow \mathrm{n} + \alpha + 17.6\,\mathrm{MeV} \tag{1}$$

However, the viability of this cycle is fundamentally bottlenecked by the supply of tritium, which in turn acts as a limiting factor for fusion energy as a whole (Pearson, Antoniazzi, and Nuttall, 2018; Khan, 2026a).

Tritium is a radioactive isotope with a short half-life ($\sim$12.3 years) and effectively no natural abundance. Current global inventories are derived primarily as by-products from heavy-water moderated fission reactors and are severely limited (Kovari et al., 2018). To contextualise this scarcity, the testing and startup of a large-scale fusion device requires a substantial initial tritium inventory of upwards of 10–12 kg, while the continuous operation of a commercial reactor is projected to consume between 10 and 100 kg of tritium annually (Lord et al., 2024). In stark contrast, current global production is optimistically capped at approximately 8 kg per year and most of this is claimed by other industries (Pearson, Antoniazzi, and Nuttall, 2018).

The only sustainable solution to this supply crisis is the in situ breeding of tritium within the fusion reactor itself. This is achieved using a 'breeder blanket': a critical reactor module wrapped around the nuclear core. While breeder blankets have historical roots in fission reactors for producing fissile isotopes (Moir, 1982; Maniscalco et al., 1984), modern fusion research utilise the neutrons emitted from the D-T process (Equation 1) to transmute lithium into tritium (see Section 2.1.1). These modules have become a key focus global fusion efforts (see Section 3.1). The critical figure of merit for this process is the Tritium Breeding Ratio (TBR), defined as the ratio of tritium produced to tritium consumed. To ensure fuel self-sufficiency and to build excess inventory for starting future reactors, the TBR must be maximised as much as reasonably possible (Sawan and Abdou, 2006).

While maximising the TBR is the primary goal, the breeder blanket is a highly constrained multi-physics environment. It must simultaneously act as the primary component for extracting high-grade thermal energy for electricity generation, whilst remaining structurally durable and safe under extreme neutron irradiation and thermal stress (Federici et al., 2019). Building upon previous neutron transport studies using homogeneous approximations (Khan, 2026a), this study aims to optimise the TBR of a discrete, heterogeneous layered blanket architecture and explores employing Machine Learning (ML) to navigate increasingly intractable physical and geometric constraints.

2. THEORY

2.1. Breeder Blanket Physics

2.1.1. Operating Principles

Because the D-T plasma yields exactly one neutron per consumption event (Equation 1), and realistic reactor geometries suffer from unavoidable neutron leakage and parasitic absorption, the breeder blanket must achieve a TBR strictly greater than unity just to maintain a continuous fuel cycle (Pearson, Antoniazzi, and Nuttall, 2018; Khan, 2026a). Furthermore, because conservation of momentum dictates that this single emitted neutron carries away four-fifths of the total fusion energy ($E_n \approx 14.1$ MeV compared to $\sim 1-2$ MeV neutrons in fission reactors (Heckrotte and Hiskes, 1971)), fusion breeder blankets must be specialised to utilise these ultra high energy neutrons.

To breed this fuel, the blanket exploits neutron capture interactions with lithium. Natural lithium consists of two isotopes, both of which undergo tritium-producing reactions. The dominant ^{6}Li reaction is exothermic and possesses a $1/v$ cross-section dependence, making it highly efficient at capturing thermalised neutrons:

$$^6\mathrm{Li} + \mathrm{n} \longrightarrow \alpha + \mathrm{t} + 4.8\,\mathrm{MeV}. \tag{2}$$

Conversely, the ^{7}Li reaction is an endothermic threshold reaction requiring fast neutrons

$$^7\mathrm{Li} + \mathrm{n} \longrightarrow \alpha + \mathrm{t} + \mathrm{n}' - 2.5\,\mathrm{MeV}. \tag{3}$$

Because the dominant ^{6}Li reaction is purely absorptive, a self sufficient neutron economy necessitates the inclusion of neutron multipliers to maintain a viable population. Elements such as lead are utilised for their substantial spallation (n, 2 n) cross-sections at high incident energies:

$$^{208}\mathrm{Pb} + \mathrm{n} \longrightarrow {}^{207}\mathrm{Pb} + 2\,\mathrm{n}' - 7.37\,\mathrm{MeV} \tag{4}$$

These reactions increase the neutron population, compensating for macroscopic breeding inefficiencies (Section 3.1). In this study, the continuous-energy nuclear data libraries utilised by OpenMC (Section 3.2) are sampled from Brown et al., 2018.

To increase the capture probability of the ^{6}Li isotope, the 14.1 MeV fast neutrons emitted by the D-T plasma must be moderated down to thermal energies. This moderation is achieved through successive elastic and inelastic scattering collisions within the blanket medium and, if using water (Khan, 2026a), internal coolants. Crucially, these scattering events are the primary mechanism by which the neutron's kinetic energy is deposited into the blanket as extractable heat.

The total thermal yield is further modified by the reaction network's kinematics: exothermic captures (Equation 2) enhance local heat deposition, whereas endothermic reactions (Equation 3, 4) inherently consume kinetic energy, reducing the net thermal yield.

Finally, to maximise both tritium breeding and energy utilisation while minimising safety risks from neutron leakage, a totally attenuated neutron flux is ideal (Khan, 2026a).

2.1.2. Trace Reactions

Structural and coolant components exhibit trace background reactions that become visible in high-resolution mesh tallies (Section 5.1). Incident fast 14.1 MeV neutrons trigger minor endothermic spallation events in elements that appear in structural materials (defined in Section 4.1). Examples include zirconium and iron:

$$^{90}\mathrm{Zr} + \mathrm{n} \longrightarrow {}^{89}\mathrm{Zr} + 2\,\mathrm{n} - 12.0\,\mathrm{MeV} \qquad (\sigma \approx 0.6 \text{ b}) \tag{5}$$

$$^{56}\mathrm{Fe} + \mathrm{n} \longrightarrow {}^{55}\mathrm{Fe} + 2\,\mathrm{n} - 11.2\,\mathrm{MeV} \qquad (\sigma \approx 0.4 \text{ b}) \tag{6}$$

In addition to multiplication, this unattenuated high-energy flux induces negligible trace tritium production via extreme threshold capture reactions in zirconium:

$$^{90}\mathrm{Zr} + \mathrm{n} \longrightarrow {}^{88}\mathrm{Y} + \mathrm{t} - 11.3\,\mathrm{MeV} \qquad (\sigma \sim 1 \text{ µb}) \tag{7}$$

Conversely, highly thermalised neutrons can undergo radiative capture with the naturally occurring deuterium in water to breed trace amounts of tritium:

$$^{2}\mathrm{H} + \mathrm{n} \longrightarrow {}^{3}\mathrm{H} + \gamma + 6.26\,\mathrm{MeV} \qquad (\text{thermal } \sigma \approx 0.5 \text{ mb}) \tag{8}$$

2.2. Neutron Transport Metrics

2.2.1. Radiation Damage and Displacements Per Atom

To accurately assess the structural viability of the breeder blanket and plasma-facing First Wall (FW), the neutronic simulation tallies radiation-induced material degradation, where the primary concern for fusion is damage by atomic displacement. The standard metric for quantifying this degradation in a crystalline lattice is Displacements Per Atom (DPA), which represents the average number of times an atom is knocked out of its equilibrium lattice site due to radiation bombardment.

When a high-energy fusion neutron collides with a target nucleus, it transfers kinetic energy, creating a Primary Knock-on Atom (PKA). If the transferred energy exceeds the lattice displacement threshold (E_d), the PKA is ejected and can trigger a secondary collision cascade. However, not all the kinetic energy transferred to the PKA contributes to atomic displacements; a significant fraction is lost to inelastic electronic excitation (ionisation) as the PKA travels through the lattice. The specific fraction of the transferred energy available to cause further elastic nuclear collisions and subsequent atomic displacements is tallied as the "Damage Energy" (T_d). Using models like the Athermic Recombination-Corrected (ARC-DPA) model (Nordlund et al., 2018) applied in this study, we can convert this into a number of displacements (N_d). To determine if a material exceeds its irradiation limits over its operational lifetime, the total number of displacements must be normalised into DPA per Full Power

Year (FPY). For a given material, reactor power and neutron spectrum we can calculate DPA/FPY (Appendix A.1) and determine a structural lifetime given prescribed DPA limits.

Alongside atomic displacements, the unattenuated high-energy fusion neutron flux induces nuclear transmutation reactions, predominantly (n, α) processes, which generate insoluble gases within the structural lattice. The accumulation of these transmutation gases is particularly critical in fusion environments. Helium atoms rapidly migrate to grain boundaries and coalesce into bubbles, leading to severe volume swelling and helium embrittlement (Trinkaus, 1985). This synergistic degradation mechanism fundamentally constrains component lifetimes. Operational embrittlement threshold limits have been drafted for fusion materials in terms of DPA, limits by Gilbert et al., 2012 are applied in this study.

2.2.2. Nuclear Heating and Energy Deposition

In addition to radiation damage, the structural and thermal-hydraulic viability of a blanket design is dictated by the spatial distribution of nuclear heating. Energy is deposited into the blanket materials through two primary mechanisms: neutron kinetic energy transfer during elastic and inelastic scattering, and the absorption of secondary gamma rays (γ-photons) produced during nuclear reactions (Kuridan, 2023). In a simulation we calculate (Appendix A.2) and tally the localised deposited kinetic energy.

2.2.3. Neutron Track Length

To quantify the neutron population within specific geometric regions of the blanket, stochastic Monte Carlo transport codes, such as OpenMC (Romano et al., 2015), utilise a track-length estimator. Rather than counting individual neutrons crossing a boundary, the simulation integrates the total distance traversed by all particle histories within a defined cell volume (Appendix A.3) to get a flux value. This accounts for neutrons that only briefly enters a cell compared to those that have a long path in the cell, to give a more physically representative tally to be tracked in neutron simulations.

2.3. ML: Bayesian Optimisation

To mitigate severe computational expense (Section 3.2), this study employs Bayesian Optimisation (BO), a sample-efficient global optimisation strategy highly suited for evaluating expensive, 'black-box' objective functions (Shahriari et al., 2016). BO is fundamentally driven by Bayes' Theorem, wherein a 'prior' belief about the objective function's landscape is iteratively updated with new simulation evidence to form a refined 'posterior' distribution. This loop relies on two coupled mathematical components: a probabilistic surrogate model and an acquisition function.

2.3.1. Surrogate Modelling via Gaussian Process Regression

A surrogate model is an approximated map of the output of a function across its full domain of input parameters, given a limited set of discrete test data to learn from. This project utilises Gaussian Process Regression (GPR) (Rasmussen and Williams, 2006) to interpolate and extrapolate the predicted value of a function based on given test data points. Unlike parametric regression models that fit a single discrete curve, a Gaussian Process (GP) defines a non-parametric probability distribution over all possible functions that could fit the observed data.

For any given coordinate in the parameter space (e.g., a specific set of blanket layer thicknesses), the GPR outputs a probability distribution characterised by two key metrics:

- **Mean Function,** $\mu(x)$**:** The expected value of the objective metric (e.g., TBR).
- **Variance,** $\sigma^2(x)$**:** The uncertainty associated with that prediction.

At locations near previously simulated data points, $\sigma(x)$ collapses to the stochastic noise floor, representing high confidence. In unexplored regions of the parameter space, $\sigma(x)$ expands. This behaviour is governed by the GP's covariance function (or kernel). The kernel mathematically dictates how information 'bleeds' between spatially adjacent points, assuming that configurations with similar geometric inputs will yield correlated neutronic outputs. GPR is particularly advantageous for Monte Carlo applications as its kernel can explicitly model the inherent statistical noise of stochastic particle tracking, preventing the optimiser from over-fitting to statistical anomalies (Snoek, Larochelle, and Adams, 2012).

2.3.2. *Acquisition Functions and the Exploration-Exploitation Trade-off*

While the GPR provides a probabilistic map of the peaks in a design space, it does not dictate the search path. This is the role of the Acquisition Function, an analytically defined, cheap-to-evaluate equation that transforms the GP's $\mu(x)$ and $\sigma(x)$ into a single scalar 'utility' score. The parameter space coordinate that maximises this utility is selected as the next candidate for testing, as the function has determined it as the most informative area to investigate. By following an optimal search path, the number of needed tests to get an accurate surrogate drastically drops.

The acquisition function must carefully balance two conflicting objectives: 'exploitation' (sampling areas with a high predicted $\mu(x)$ to refine known local maxima) and 'exploration' (sampling areas with high $\sigma(x)$ to discover untracked global maxima).

While several commonly employed acquisition functions exist (Appendix A.4), the Upper Confidence Bound (UCB) acquisition function (Srinivas et al., 2012) has a simple but powerful implementation that is directly tunable:

$$UCB(x) = \mu(x) + \sqrt{\beta} \cdot \sigma(x) \tag{9}$$

UCB evaluates the absolute best-case score for any given point given the sum of its predicted $\mu(x)$ and $\sigma(x)$. The hyperparameter β acts as an explicit control over the algorithm's optimism. By setting β to a sufficiently high value, the algorithm is forced to aggressively explore highly uncertain regions (demonstrated in Section 4.3.1).

3. LITERATURE REVIEW

3.1. Breeder Blanket Architectures

Though Section 2.1.1 defined the theoretical TBR > 1 requirement, literature commonly cites a minimum of 1.05 for practical breeding to account for parasitic neutron losses and extraction inefficiency (Tanabe, 2017; Hernández and Pereslavtsev, 2018; Khan, 2026a). Fusion projects across the globe explore diverse material paradigms—ranging from solid ceramics in ITER (Federici et al., 2019) and liquid metal concepts in the China Fusion Engineering Test Reactor (CFETR) (Zhuang, Li, Jian, et al., 2019), to highly compact liquid immersion variants in the Spherical Tokamak for Energy Production (STEP) and the Affordable, Robust, Compact (ARC) reactor (Lord et al., 2024; Sorbom et al., 2015)—that were explored in Khan, 2026a. However, evaluating breeder blankets opens many questions in intra-blanket structure and design. Practical blanket modules cannot operate as idealised, homogeneous mixtures; they require complex, layered internal cooling channels and structural separators to ensure thermo-mechanical stability.

A review of current mature conceptual designs, such as the Water-Cooled Lithium-Lead (WCLL) and Helium-Cooled Pebble Bed (HCPB), demonstrates that coolants and their associated structural containment typically occupy 30%–40% of the total blanket volume (Del Nevo et al., 2019). The coolant distribution is heavily skewed towards the front of the blanket for optimum heat extraction. Current literature explores a spectrum of cooling architectures, ranging from bulk liquid immersion variants or coolant/self-cooled layers (Lord et al., 2024; Sorbom et al., 2015; Segantin et al., 2020; Kirillov et al., 1998) to designs that rely on discrete networks of internal cooling channels (Federici et al., 2019; Zhuang, Li, Jian, et al., 2019; D. W. Lee et al., 2012; Qu et al., 2020). To prevent structural failure under extreme loads, these internal cooling networks are typically integrated within Reduced-Activation Ferritic-Martensitic (RAFM) steels, notably EUROFER97 (Stornelli et al., 2024), and are frequently augmented by sacrificial tungsten armour (You et al., 2016)

The FW endures the most severe environmental stressors in the reactor; demonstration power plant (DEMO) concepts state they must mitigate surface heat fluxes often exceeding 1.0 MW/m^2 alongside peak neutron wall loadings of $\sim$ 2.0 MW/m^2 (You et al., 2016). These extreme thermal and neutronic loads strictly dictate component lifetime propositions. While ITER Test Blanket Modules are designed to survive experimental campaigns with cumulative radiation damage below 3 DPA, commercial power plants will subject blanket structures to 20–50 DPA per FPY (Federici et al., 2017). This necessitates a modular design approach, with structural replacement anticipated every 2–5 years to prevent catastrophic embrittlement (Boccaccini et al., 2014).

However, the integration of extensive structural and cooling networks inherently degrades the global neutron economy. Blanket design ceases to be a purely neutronic exercise and becomes a highly constrained multi-physics trade-off. As demonstrated in recent optimisation studies (Qiu, Lu, Boccaccini, et al., 2018), researchers must continually refine spatial layer configurations to maximise the primary objectives, TBR and high-grade heat extraction, without exceeding the strict material degradation and operational limits imposed by the heterogeneous environment.

3.2. Computational Load in Nuclear Neutronics

High-fidelity neutronic evaluation necessitates stochastic Monte Carlo transport methods (Kalos and Whitlock, 2008), using industry standard codes such as OpenMC (Romano et al., 2015). Achieving statistically significant tallies in deeply shielded or structurally intricate regions imposes a severe computational burden. As explicitly quantified in recent fusion power plant studies by Warmer et al., 2024, sufficiently accurate Monte Carlo neutronics simulations routinely require 10^8–10^9 particle histories, directly leading to several thousand CPU-hours on dedicated supercomputing clusters per run. Consequently, executing traditional exhaustive grid-search optimisations across highly dimensional design spaces results in exponential scaling that is functionally intractable. Humphrey et al., 2023 highlights, in their evaluation of fusion component design, the immense computational cost of Monte Carlo transport precludes rapid iterative optimisation, strictly limiting traditional parametric sweeps to low-dimensional or heavily homogenised approximations.

3.3. Machine Learning and Surrogate Modelling in Physics

To circumvent the severe computational limitations of high-fidelity transport codes, the broader computational physics community has increasingly adopted surrogate modelling and sequential learning techniques. Methods such as BO and GPR (Section 2.3) are distinctly suited to resolving this specific bottleneck (Rasmussen and Williams, 2006). Rather than exhaustively evaluating an entire design space, these algorithms intelligently sample a small fraction of the data points required by a traditional grid search. This targeted approach yields a robust understanding of the optimal parameter landscape while drastically reducing the required number of computationally expensive simulations (Emmerich, Giannakoglou, and Naujoks, 2006; Shahriari et al., 2016).

The translation of this machine learning paradigm into nuclear fusion neutronics is a rapidly emerging, cutting-edge frontier. For instance, Mánek et al., 2023 recently established the efficacy of using ML surrogates to efficiently regress the TBR from heavy Monte Carlo datasets. The explicit application of BO to optimise complex breeder blanket architectures is just now being proven as a legitimate, highly competitive route. Concurrent, state-of-the-art work by Humphrey, Brooks, Mungale, et al., 2026 at the UK Atomic Energy Authority (UKAEA) demonstrates the viability of deploying Bayesian sequential learning to navigate the parameter space of an OpenMC neutronic model. The recent emergence of such closely related approaches suggests a potential paradigm shift and validates the timeliness, originality, and necessity of the methodology developed in this study.

4. METHODOLOGY

4.1. Simulation Pipeline Summary

The neutronic performance of our breeder blanket geometries is evaluated using a coupled computational workflow. The 3D parametric Computer Aided Design (CAD) models of the tokamak reactor are generated using the Python-based framework Paramak (Shimwell et al., 2021). To ensure computational efficiency while maintaining physical accuracy, the simulation domain comprises a 90° toroidal sector with periodic boundary conditions applied to the bounding surfaces. These Constructive Solid Geometry (CSG) geometries are exported as Direct Accelerated Geometry Monte Carlo (DAGMC) unstructured meshes, which allow for efficient ray-tracing directly on the CAD surface representations without the need for voxelisation (Romano et al., 2015). Stochastic neutron transport simulations are subsequently executed upon these meshes using OpenMC (Romano et al., 2015). To ensure statistical robustness and minimise relative error below 0.5%, simulations utilised high particle counts ranging between 10^5 and 10^7 neutrons per run.

Based on the optimal configurations identified in Khan, 2026a, the material definitions for this architecture are explicitly fixed: the FW utilises a zirconium alloy, the breeder is a lithium-lead eutectic ($Li_{17}Pb_{83}$) with 40% enriched ^6Li, and the coolant is pressurised water (modelled at 600 K and 15.5 MPa). EUROFER97 was selected for the internal structural separators, rear wall, and central solenoid, reflecting industry standards for robust, reduced-activation ferritic-martensitic steels. The

14.1 MeV Muir-distributed ring source utilised in Khan, 2026a has been upgraded to a volumetric body source with a denser core; the default OpenMC parameters are specified in Fausser, Li Puma, and Gabriel, 2012. A baseline reactor thermal power of 500 MW was assumed to scale DPA/FPY (Section 2.2.1), reflecting the planned operational output of ITER (Aymar, Barabaschi, and Shimomura, 2002).

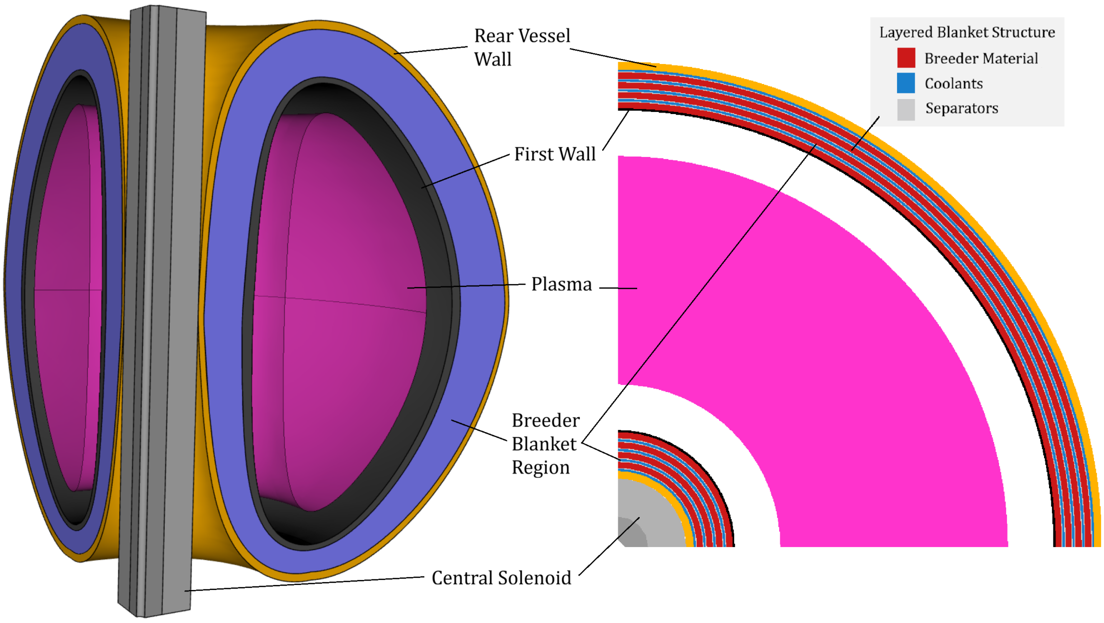


Figure 1: Representative geometry of the simplified Tokamak model generated via Paramak. Included is an example breakdown of the layered blanket, only the coolant and breeder thicknesses are adjusted in this study.

A comprehensive exposition of this foundational workflow was established in the first stage of this study (Khan, 2026a). Building upon those established homogeneous baseline models, this current study transitions to a discrete, heterogeneous layered blanket architecture. We use 4 breeder layers and 4 coolant layers, with separators between each breeder-coolant interface (see Figure 1). This addresses the core unrealistic homogenous approximation of that study, establishing a key step towards more realistic breeder blanket module requirements and challenges. Paramak's parameterised framework limits the level of heterogeneity to systems with azimuthal symmetry (such as our layered model). Note that while other studies have simulated detailed asymmetric coolant channel designs, related 'bulk-coolant' designs do exist (Section 3.1). Despite approximations and deviations compared to specific real implementations, we proceed as there are key advantages to this approach:

- **Conservative heterogeneity:** Applying this model allows for the majority of heterogeneous effects to be modelled. Importantly, in real implementations streaming effects allow some neutrons to bypass the interactions between coolants and separators. The perfectly layered model prevents this, forcing heterogenous material paths. Given such heterogeneity negatively affects performance (demonstrated in Section 5.1), the layered model notably becomes a conservative estimate of blanket performance. In an immature field where specific designs are undecided, a conservative study can be informative.
- **Parametrisation:** A layered model is easily parameterised into a set of thickness variables for each layer. Comprehensively exploring and optimising even this 8-layer model required complex techniques that became a majority component of this project (see Section 4.3). Using a more detailed model necessitates a greater number of parameters and thus exploring higher dimensional spaces. This is non-trivial to optimise, it is instructional to begin with a lower dimensional parameterised model before scaling.

4.2. Neutronic Analysis

4.2.1. Mechanistic Analysis of the Layered Blanket

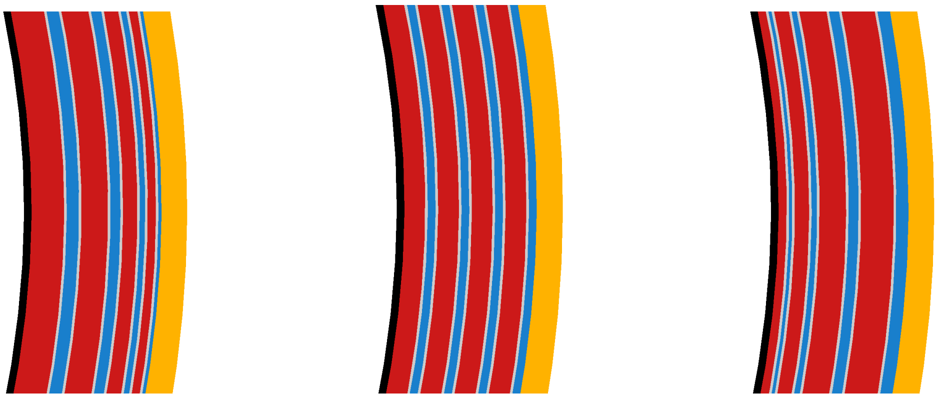
Initial parametric thickness sweeps were conducted, keeping all other material proportions constant, to isolate the effect of heterogeneity. We included tallying of individual layers to understand which were contributing and impactful to global results (Section 5.1).

Mechanistic understanding of performance necessitates localised tracking of several neutron transport parameters (Section 2.2). To generate 3D spatial distributions, regular Cartesian mesh tallies were overlaid onto a high-resolution 3D bounding box mesh defined across the reactor. Within this grid, six specific reaction metrics (Section 2.2) were scored independently using OpenMC's standard tallies: neutron flux, scattering density, tritium production, neutron multiplication, nuclear heating, and damage energy. The raw outputs from these mesh tallies were subsequently processed into a 2D poloidal slice of the reactor and visualised as heat maps (Section 5.1).

4.2.2. Manual Thickness Distribution Analysis

To systematically investigate the effect of radial material distribution within the heterogeneous blanket, a constrained parametric approach was initially required. The full layered architecture contains eight independent geometric variables (comprising four discrete breeder layers and four coolant layers). Exhaustively exploring this 8-dimensional design space via traditional manual or grid-search methods is computationally intractable (Section 3.2).

To reduce the dimensionality of the problem while maintaining physical relevance, a linear gradient parameter, ζ, was introduced as a heuristic constraint (see Figure 2). This parameter governs the relative thickness of the layers across the blanket's depth (while holding seperator thickness constant), providing a human-intuitive method to observe spatial trends. In Section 5.1 we aggregate how neutronic parameters vary under the sweep of ζ, bound to $|\zeta| < 0.35$ from meshing constraints in thin layers.



(a) $\zeta < 0$ **(b)** $\zeta = 0$ **(c)** $\zeta > 0$

Figure 2: Cross-sectional visualisations of the breeder blanket layers illustrating the effect of the linear gradient parameter, ζ. This parameter controls the radial distribution of layer thicknesses: (a) a forward-weighted geometry shifting volume towards the FW, (b) the baseline uniform thickness distribution, and (c) a rear-weighted geometry shifting volume deeper into the blanket.

While this constrained, single-variable approach yields valuable insights into the fundamental physics of the blanket, it inherently restricts the exploration of the broader design space. The intractability of manual parametrisation in identifying true global optima across an unconstrained 8-dimensional parameter space necessitate the use of advanced computational techniques. Consequently, the study transitions to the application of ML algorithms (Section 4.3) to navigate this complex parameter space efficiently.

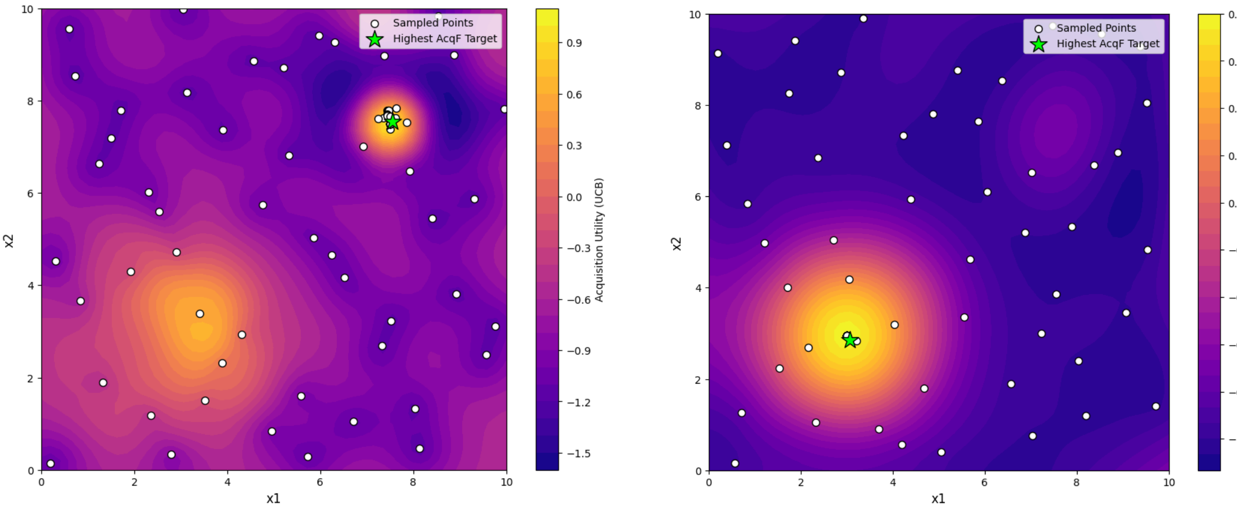
4.3. ML Optimisation

4.3.1. ML Algorithm

To navigate the computationally expensive 8-dimensional design space effectively, Single-Objective Bayesian Optimisation was employed with a Gaussian Process Regressor was utilised as the surrogate model (SOBO-GPR) (Section 2.3). The optimisation pipeline was developed using the Ax framework (Olson et al., 2025), leveraging BoTorch (Balandat et al., 2019) and PyTorch (Paszke et al., 2019) as the underlying computational engines. This scalable, N-dimensional architecture is highly sample-efficient, capable of converging upon optimal configurations within $N < 150$ trials (tested in spaces up to 8 dimensions). The efficiency makes it ideal for optimising our computationally intensive physical simulations. Furthermore, the objective function and hyperparameters are designed to be modularly configurable, with automated generation of surrogate parameter surfaces and optimisation progress diagnostics. The complete source code and optimisation logs are publicly available (Khan, 2026b).

The objective function was defined by wrapping the discrete parametric OpenMC neutronics simulation into an automated evaluation loop. This function takes the eight blanket layer thicknesses as continuous input parameters (bounds defined in Section 4.3.3) and returns the global TBR as the single scalar objective to be maximised. The computational overhead of the ML solver was negligible ($\approx 20\,\mathrm{s}$ for $N = 150$ trials); thus, the total runtime was bounded entirely by the neutronics simulation, which required $\approx 60\,\mathrm{s}$ per trial on a standard desktop workstation. Because the optimiser was constrained to a maximum of $N = 150$ trials, all ML optimisation studies were completed in under 3 hours.

For this study, the UCB acquisition function (Equation 9) was selected for its explicit, scalar control over the exploration-exploitation trade-off via the hyperparameter β. Properly tuning this parameter is vital for a high-dimensional physical system where multiple local maxima may exist. Figure 3 illustrates a 2D projection of the acquisition utility landscape used to benchmark the model's behaviour.



(a) Explorative tuning ($\beta = 1,000,000$)

(b) Greedy tuning ($\beta = 1$)

Figure 3: Comparison of the Upper Confidence Bound (UCB) acquisition function behaviour under different exploration parameters (β).

As demonstrated in Figure 3b, setting a low exploration weight ($\beta = 1$) results in a overly greedy search. The algorithm heavily exploits a known local optimum, entirely missing a secondary, hidden peak within the design space. Conversely, Figure 3a illustrates that a significantly higher parameter ($\beta = 1,000,000$) forces the model to prioritise high-uncertainty regions, successfully mapping the broader landscape and identifying the hidden peak.

Given the complexity and scale of the layered blanket parameter space, wider exploration is imperative. Consequently, the UCB acquisition function was configured with this highly explorative $\beta = 1,000,000$ value to ensure robust, global mapping of the design space before narrowing in on the optimal layer thicknesses.

4.3.2. Testing, Tuning & Developing the Optimiser

Before applying the Bayesian optimiser to the high-dimensional neutronic blanket model it was evaluated against a suite of notoriously difficult mathematical landscapes, including the Rastrigin (Rastrigin, 1974), Schwefel (Schwefel, 1981), Rosenbrock (Rosenbrock, 1960), and Gramacy & Lee (Gramacy and H. K. Lee, 2012) functions, to rigorously benchmark the optimiser's robustness against stagnation. These functions were selected for their diverse pathological traits; such as high-frequency noise, deceptive local minima, and non-symmetric valleys. While 1D tests validate basic functionality, scaling the benchmark to 2D surfaces is necessary to ensure the algorithm does not fail as dimensionality increases. These functions were scaled incrementally from 1D up to 4D to ensure the ϵ-greedy and peak-hunting mechanisms remained effective as the search space expanded exponentially.

A notable distinction highlighted here is the difference between the 'Actual Best Observed' (the highest discrete point successfully sampled during the simulation) and the 'Best Model Prediction' (the theoretical peak calculated by the surrogate model). While finding the absolute optimal TBR is the primary engineering goal, the full continuous surrogate model is arguably the most valuable output for a physicist, providing physical intuition regarding how variables interact across the entire domain.

During these tests, it became apparent that standard Upper Confidence Bound (UCB) acquisition, even with a high exploration parameter, could occasionally stagnate in discovered maxima. To prevent this, three bespoke algorithmic enhancements were introduced:

- **Interleaved random injection:** An ϵ-greedy approach was implemented, using a defined probability parameter to periodically force a completely random sample, ensuring exploration in areas the UCB function might otherwise ignore. As SOBO-GPR is able to optimise local peaks quickly, we found setting the probability to $\epsilon = 0.5$ optimal. This was further informed by finding that UCB acquisition 'randomly explored' more in higher dimensions as the space gets exponentially larger.
- **Duplicate avoidance:** Once stuck in a local minimum, the raw BO algorithm would commonly repeatedly test the peak of that minima. Our optimiser was constrained to reject repeated coordinates, forcing continuous spatial exploration and preventing wasted computational resources on identical measurements.
- **Multimodal surrogate exploitation:** Every n_{hunt} trials, the algorithm actively scans the surrogate model to identify the top five predicted peaks. It then forces a sample at the highest peak that remains locally unexplored. In our trials we found $n_{hunt} = 10$ sufficient.

In successfully navigating challenging mathematical landscapes and recreating them effectively in the surrogate, as demonstrated throughout Figure 4, the optimiser's robust resistance to local minima was established.

Figure 4a exemplifies that comprehensive mapping and understanding of 1D test functions is easily achievable with a tuned base SOBO-GPR algorithm. However, Figure 4c and 4d demonstrate the utility of the bespoke enhancements. The algorithm was able to handle the difficult hidden peak problem (where a narrower, slightly taller peak is hidden among a wide, clear peak) with confidence, utilising the duplicate avoidance and multi-modal exploitation algorithms to avoid getting trapped. Trialling it further with several smaller peaks in the 2D schwefel indicates that the increased exploration allows for not just a successful optimisation, but a well informed and accurate final surrogate.

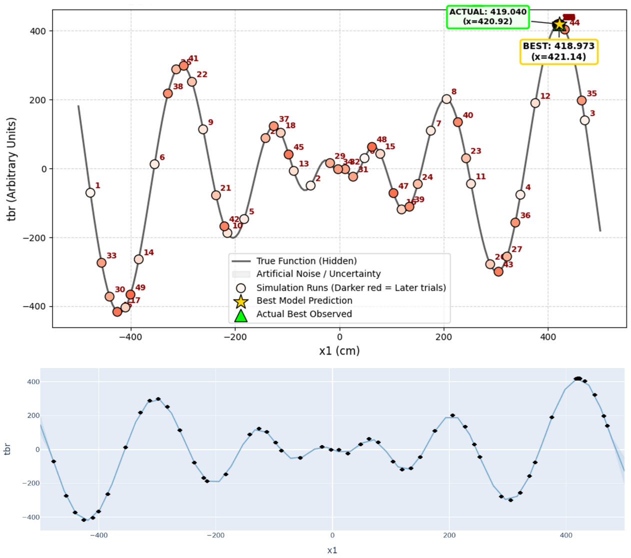
Macroscopic physics optimisation, like breeder blanket structure, is expected to feature broader, continuous global physical trends rather than sharp, discontinuous mathematical spikes. As illustrated in Figure 4b, the model efficiently isolates and solves for broad overall trends, but exhibited less sensitivity to small-length-scale variations. Therefore, while the algorithm is designed to be 'over-prepared' to handle small localised peaks, there remains a risk of optimiser local peak entrapment which could yield a marginally suboptimal result.

4.3.3. ML Optimisation Studies for TBR-Thickness

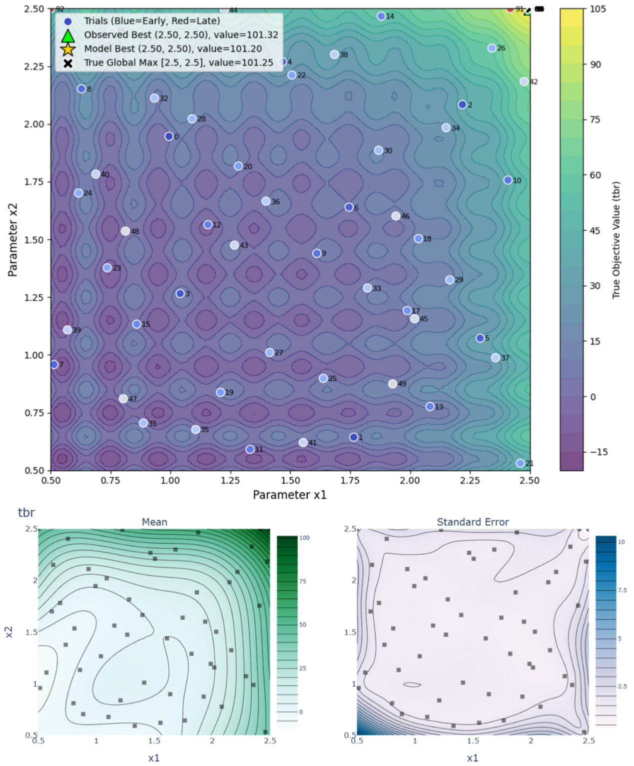
To formally bound the Bayesian optimiser against unrealistic values and meshing errors, dimensional limits were imposed on the blanket components based on engineering constraints. Discrete breeder layers were restricted to thicknesses between 3.0 cm and 16.0 cm, while coolant layers were bounded between 1.5 cm and 6.0 cm.

The model was applied on the parameterised simulation function, with the single optimisation objective of maximising TBR given layer thicknesses. Throughout testing, corner/boundary solutions experiencing binding constraints were common when no additional constraints were applied. Therefore, in the final set of studies we present several optimisation runs in which we applied additional parameter constraints. This emulates practical restrictions for more physically informative results.

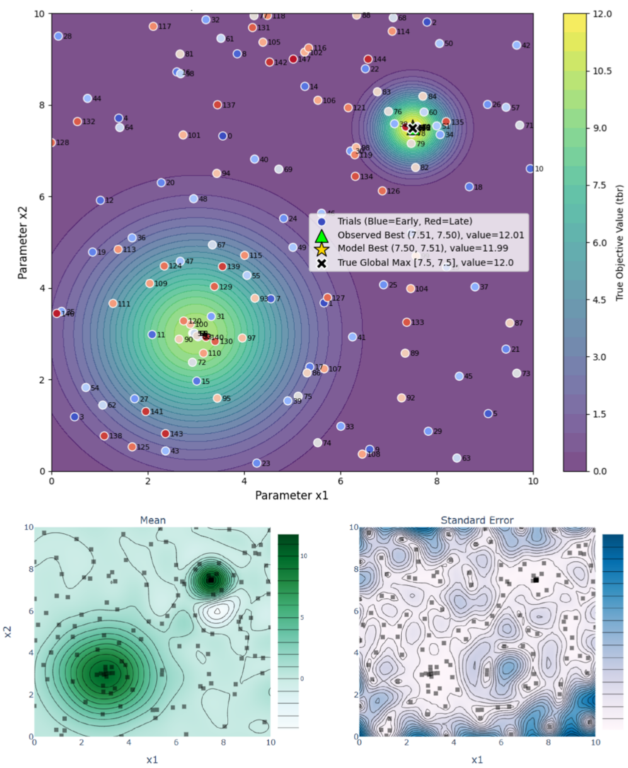
- **8D:** The model is free to adjust all 8 layers.
- **8D Constrained:** A strict total blanket thickness maximum of 44.0 cm.
- **4D Breeder Constrained:** Breeder layers were limited to a collective 32.0 cm maximum thickness, with all coolant layers locked at their 1.5 cm minimum limit.
- **4D Coolant Constrained:** Coolant layers were required to meet a minimum collective thickness of 12.0 cm, with breeder layers locked at 8.0 cm each.



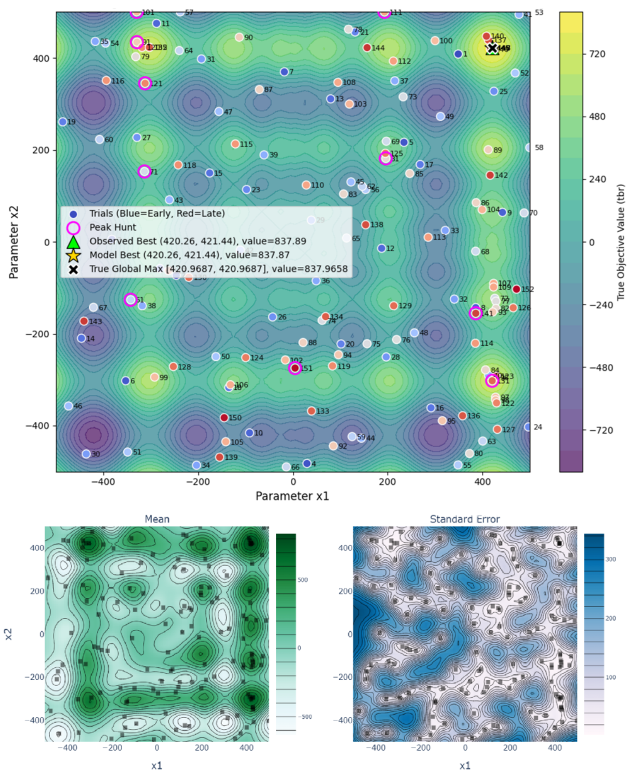
(a) 1D multi-modal Schwefel function test



(b) 2D Gramacy & Lee function test



(c) 2D Hidden Peak function test



(d) 2D multi-modal Schwefel function test

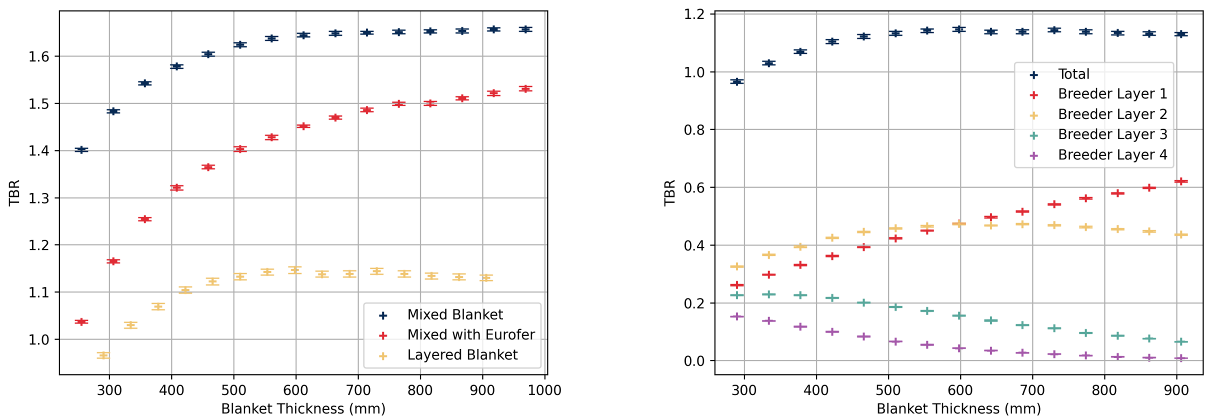
Figure 4: Search path of the ML algorithm across various landscapes, with the corresponding final surrogate plotted underneath. In (a) model uncertainty is plotted but too small to be visible. In (d) Pink markers annotate where a secondary “peak hunt” is triggered. (a, b, d) used $N = 150$ trials, while (c) used $N = 100$. All tests utilised $N_{sobol} = 50$ and UCB $\beta = 1,000,000$. Bespoke algorithm enhancements were not enabled for (a) and (b). Enhancements in (c) and (d) used $\epsilon = 0.75$, and $n_{hunt} = 10$.

On receipt of the final ML models and optimised configurations (Section 5.2), an additional manual 'Binding' configuration was generated to establish a useful comparison for the 8D constrained study.

5. RESULTS

5.1. Neutronic Behaviour of Heterogeneous Architectures

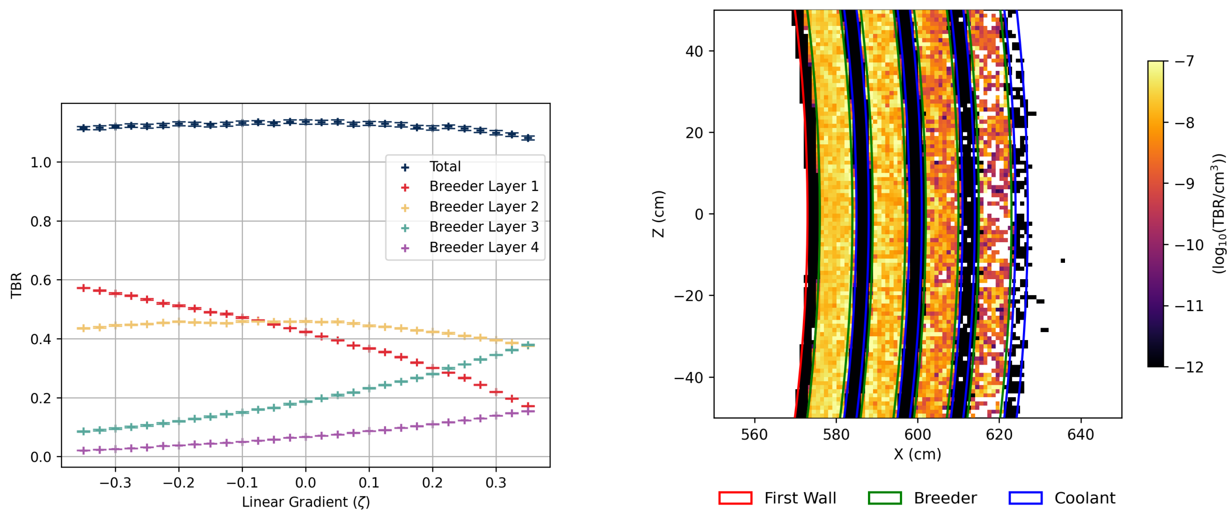
Figure 5a presents the global TBR across different modelling approaches. The data shows an initial reduction in the calculated TBR when the required structural separator material (EUROFER97) is integrated into the homogeneous mixture. A further, more substantial decrease in the global TBR is observed when this exact same material volume is heterogeneously separated into distinct breeder, coolant, and structural layers. Note all tallies in this report are given in units per source neutron.



(a) TBR vs. thickness across modelling approximations

(b) Layer-wise TBR contribution vs. thickness

Figure 5: (a) Comparison of TBR for different structures: homogenous (as in Khan, 2026a), mixed in seperator material and the $\zeta = 0$ layered model from Section 4.2. (b) Plots the layer-wise TBR breeder layer contributions for latter.

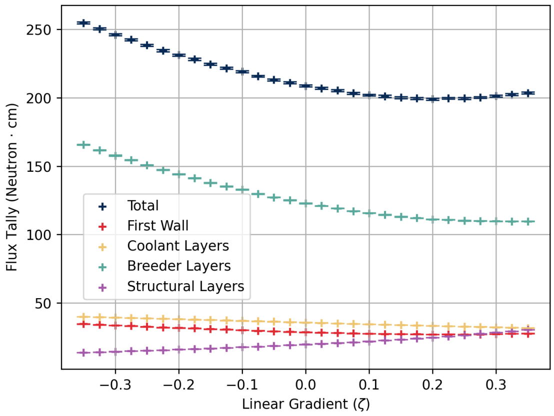


(a) TBR vs ζ

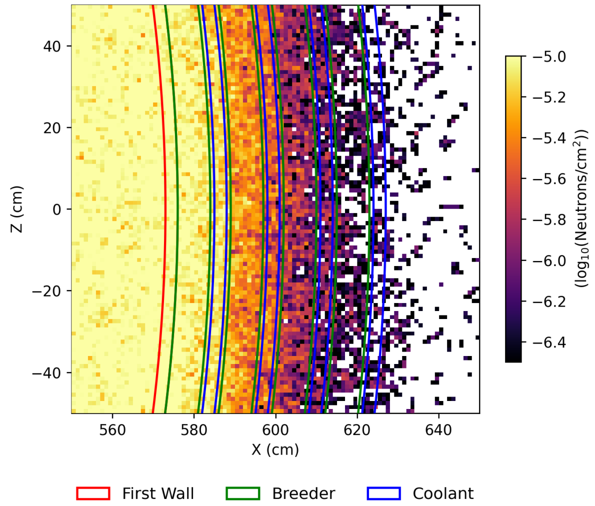
(b) TBR Reaction Density ($\zeta = 0$)

Figure 6: (a) Global and layer-specific TBR performance, paired with a heatmap of breeding reactions (b).

In evaluating heterogeneous performance (Figure 5b), breeder and coolant depths (not separators) were uniformly scaled. Global TBR rapidly saturates at ≈ 500 mm. Beyond this threshold, total



(a) Total Flux vs ζ



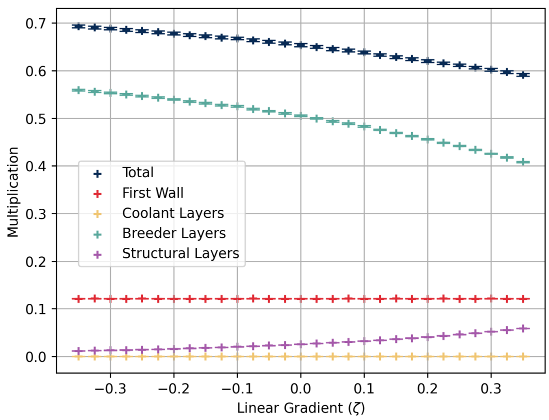
(b) Fluence Map ($\zeta = 0$)

Figure 7: (a) Neutron track length tally within reactor components, paired with a heatmap of neutron fluence depletion through the reactor wall (b).

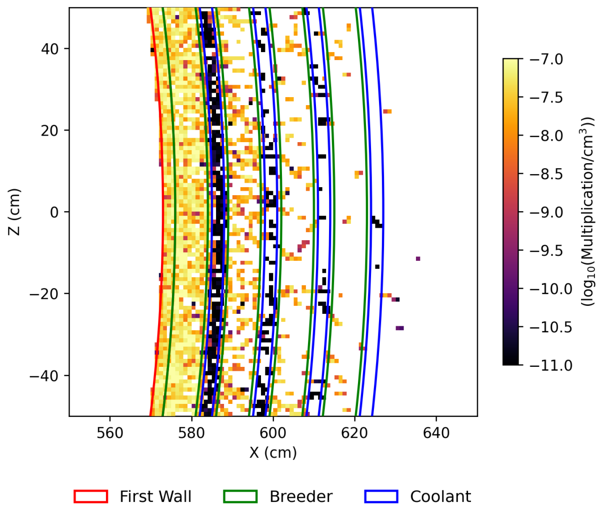
production remains constant, yet internal breeding shifts: the contribution from layer 1 rises, layer 2 stabilises, and deeper layers (3 and 4) contribute progressively less.

The global TBR remains relatively constant across the linear gradient parameter (ζ), but the spatial distribution of tritium production shifts significantly between the discrete layers (Figure 6a). The 2D reaction density map (Figure 6b) demonstrates that breeding is asymmetrically focused near the plasma-facing boundary. Notably, at a uniform thickness distribution ($\zeta = 0$), the tritium bred in the second breeder layer actually exceeds that of the first.

The volume-integrated flux tally (Figure 7a) indicates there is a greater proportion of neutrons that survive longer and penetrate further into the blanket in forward-weighted ($\zeta < 0$) configurations. The fluence map (Figure 7b) reveals that the incident neutron flux is completely attenuated, 'saturating' the blanket. Additionally, a minor step-like attenuation pattern is visible immediately following the discrete coolant layers.



(a) Multiplication vs ζ

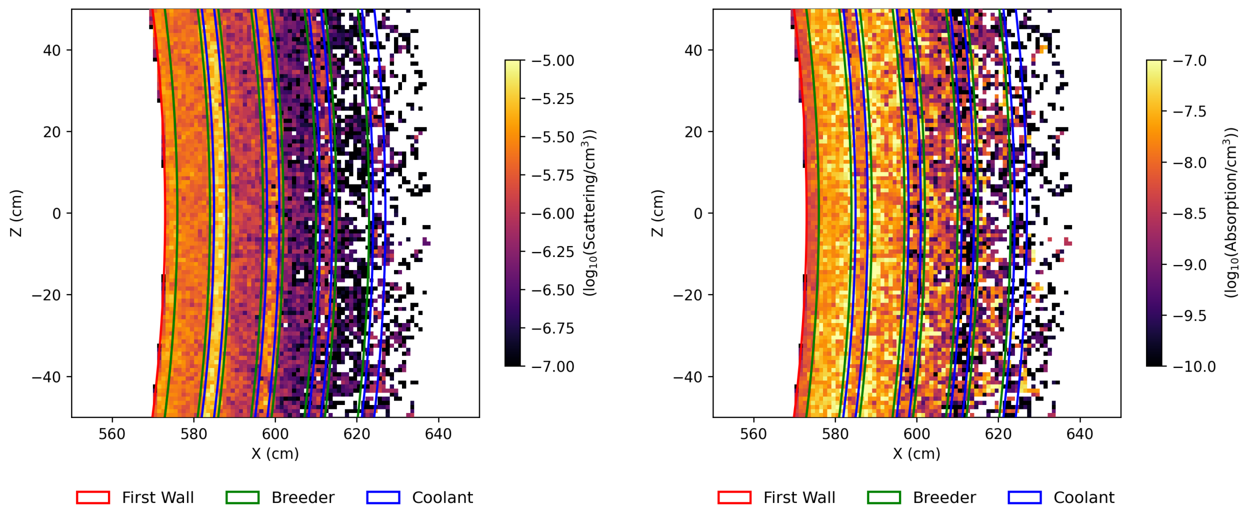


(b) Multiplication Density ($\zeta = 0$)

Figure 8: (a) Total neutron multiplication rate per source neutron split between reactor component, alongside the spatial multiplication concentration heatmap (b).

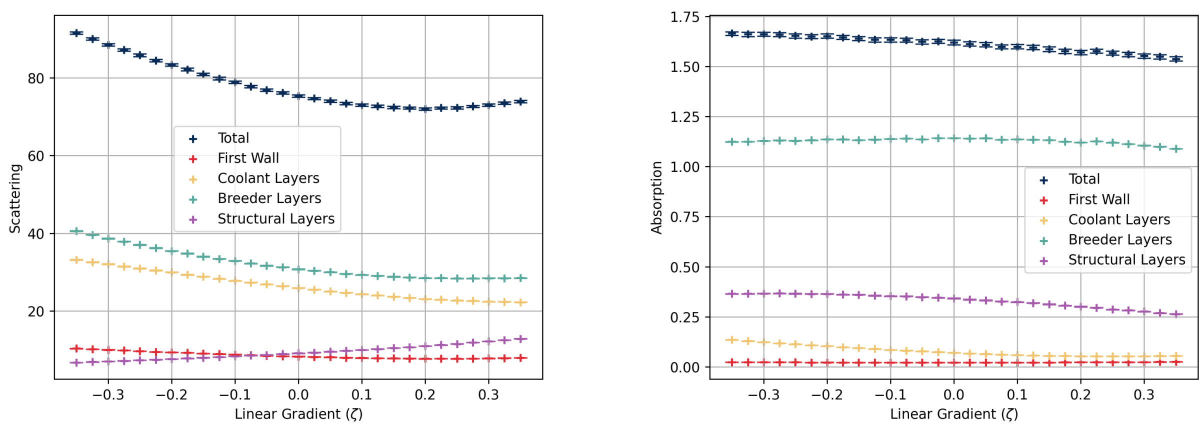
Figure 8a shows that total neutron multiplication decreases as ζ increases, yielding the highest multiplication rates in forward-weighted configurations. The spatial density map (Figure 8b) highlights that these spallation events are heavily localised within the first two breeder layers.

Neutron scattering events exhibit distinct, localised peaks within the water coolant layers (Figure 9a). Figure 9b shows strong absorption within the breeder material, weak absorption in the FW, slightly stronger interactions in the coolant, and pronounced, bright absorption peaks within the structural separators. Total absorption remains relatively constant across all ζ configurations, averaging approximately 70% within the breeder, 20% in the separators, and 6% in the coolant. Figures 10a and 10b demonstrate that as ζ increases, absorption interactions from the separators decrease but the scatterings *increase*. Although the variable neutron population from variable multiplication across ζ affected the other tallies, the percentage change in interactions from the separators are clear.



(a) Scattering Density ($\zeta = 0$) **(b)** Absorption Density ($\zeta = 0$)

Figure 9: Heat maps of scattering (a) and total absorption interactions (b) in the reactor wall. Absorption tally records all absorptions, including breeding, spallation and parasitic reactions.



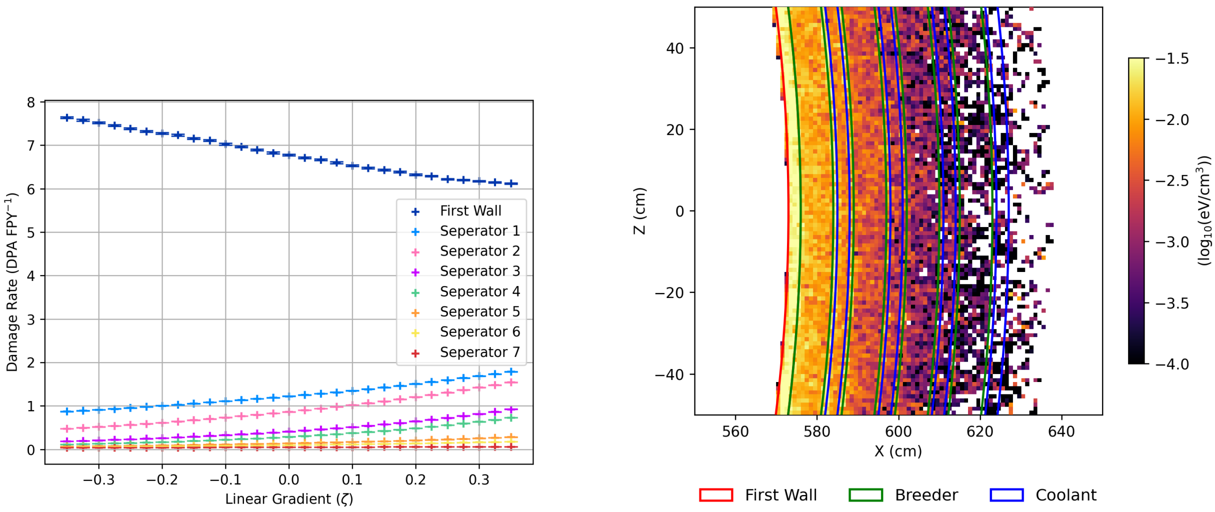
(a) Scattering vs ζ **(b)** Absorption vs ζ

Figure 10: Total volume-integrated neutron scattering (a) and absorption (b) events tracked against ζ.

Total nuclear heating also remained relatively constant against ζ at roughly 8 MeV per source neutron across all evaluated geometries. Of the total energy deposition (for these steady-state simulations), approximately 75.0% is deposited in the breeder layers and 19% in the coolant layers. There is greater heating and energy deposition per unit volume in the coolant.

Finally, Figure 11a tracks structural degradation, showing that the DPA experienced by the FW increases in forward-weighted ($\zeta < 0$) configurations . Damage energy is correspondingly correlated to DPA trends (Section 2.2.1). The spatial distribution of damage energy (Figure 11b) is heavily concentrated on the extreme plasma-facing layers. In contrast, the DPA sustained by the internal structural separators rises in rear-weighted ($\zeta > 0$) configurations.

For supplementary data on proportions, ζ trends and other results stated, see Appendix B.



(a) DPA Rate vs ζ **(b)** Damage Energy Density ($\zeta = 0$)

Figure 11: Radiation damage rate metrics in each solid layer (a) alongside the spatial profile of damage energy deposition (b).

5.2. ML Optimisation Findings

The optimal blanket configurations identified by the Bayesian optimiser under various dimensional constraints are presented in Figure 12.

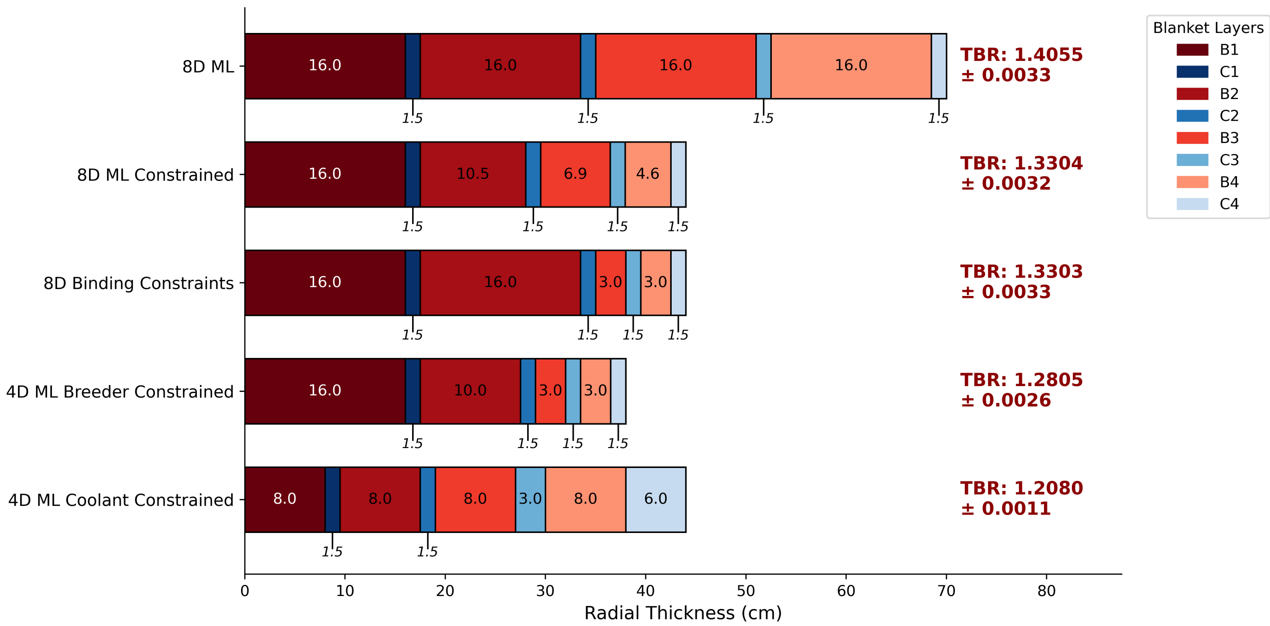


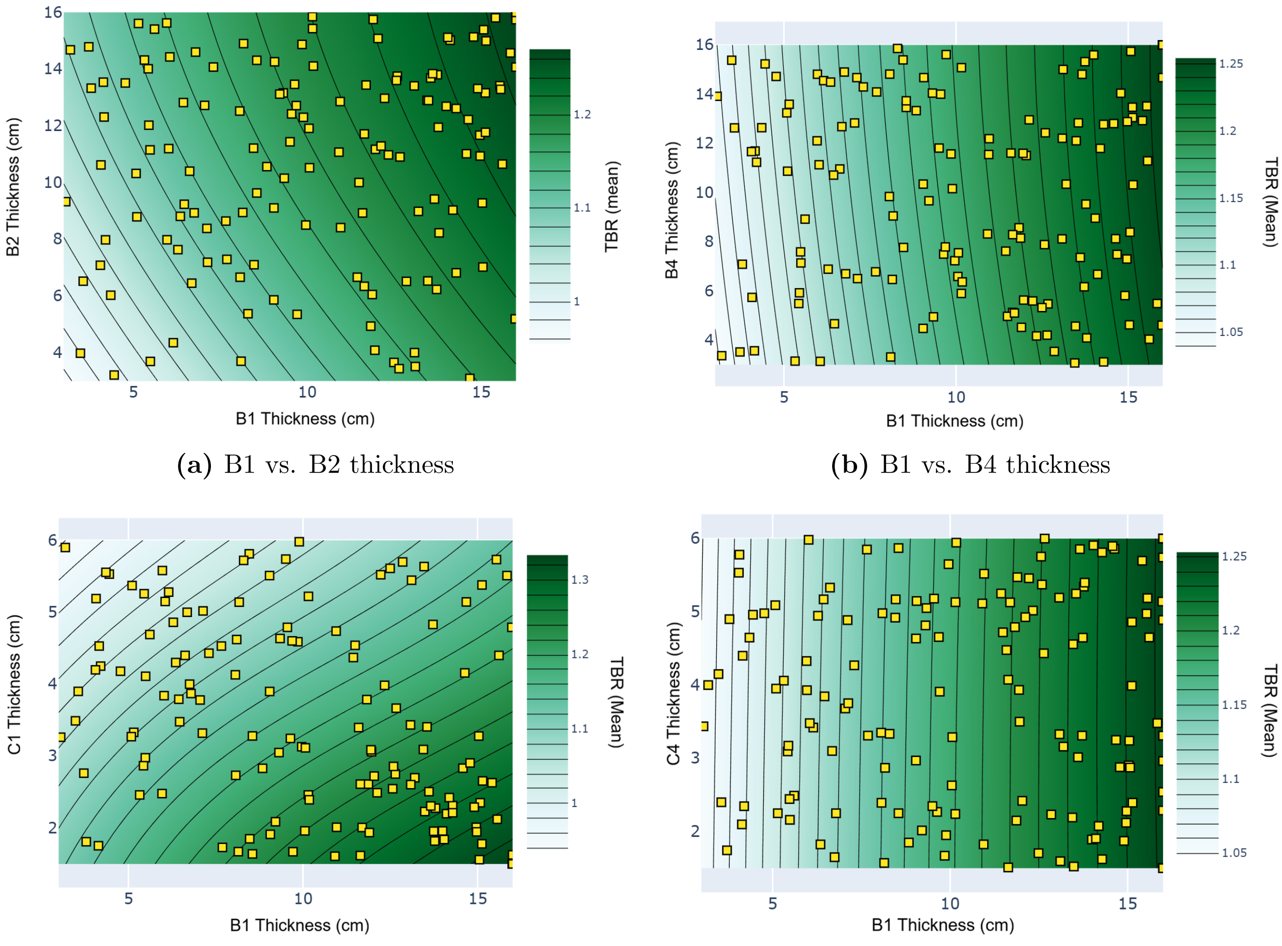
Figure 12: Radial thickness profiles and resulting global TBR for the ML optimised TBR-thickness studies detailed in Section 4.3.3. Associated TBR values are annotated alongside their standard statistical uncertainty. $N = 100$ trials were used for all except for unconstrained 8D using $N = 150$, where these include $N_{sobol} = 50$. Hyperparameters $\epsilon = 0.5$ and $\beta = 1,000,000$ were tuned for these studies.

The fully unconstrained 8D configuration achieved the highest overall TBR of 1.4055 by pushing all parameters to their 'binding limits'. Binding constraints / limits in our model is defined as when the breeder layers are maximised to their 16.0 cm upper bound while minimising the interleaving coolants to 1.5 cm. This prioritises inner layers and continues radially outward until a limit is met. The partially constrained 4D configurations exhibited lower overall breeding performance. Restricting the optimiser strictly to the maximum breeder budget or the minimum coolant budget resulted in consecutive decreases in the maximum achievable breeding ratio. Both resulted in binding conditions (Section 4.3.3), where the 4D coolant study pushed all the coolant volume to the outermost layers as much as physically permissible.

When restricted to the 44.0 cm total thickness budget ('8D ML Constrained' in Figure 12), the optimiser contrastingly identified an 'exponentially decaying' configuration where coolants were again minimised but the breeder layer thickness values instead decayed radially outward, yielding a TBR of 1.3304. By comparison, the manually defined binding constraint configuration yielded a statistically matching TBR of 1.3303.

For clarity in evaluating the high-dimensional space, the discrete blanket layers are denoted sequentially from the plasma-facing FW outwards: breeder layers are labelled B1 through B4, and their adjacent coolant layers are labelled C1 through C4.

To visualise the parameter landscape, 2D projections of the unconstrained 8D surrogate model were generated (Figure 13). These slices plot the thickness of the innermost breeder layer (B1), which exhibited the strongest single-variable influence on the TBR, against other selected layers.



(a) B1 vs. B2 thickness

(b) B1 vs. B4 thickness

(c) B1 vs. C1 thickness

(d) B1 vs. C4 thickness

Figure 13: 2D contour projections of the 8D surrogate model from the unconstrained optimisation run. Each panel plots the thickness of the innermost breeder layer (B1) against a secondary layer (B2, B4, C1, or C4), with all non-plotted dimensions held constant at their optimal values.

The gradients within these projections directly illustrate parameter sensitivity. Highly slanted contour lines, such as B1 against B2 (Figure 13a) or C1 (Figure 13c), indicate that the TBR is sensitive to variations in both plotted variables. Here the predicted maximum TBR consistently isolates at the boundary corners, favouring maximised breeder thicknesses and minimised coolant thicknesses.

Conversely, projections mapping B1 against the outermost layers, such as B4 (Figure 13b) or C4 (Figure 13d), display nearly vertical contour lines. This indicates that the global TBR is highly insensitive to dimensional changes in the outermost blanket layers compared to the innermost layers.

6. DISCUSSION

6.1. Physical Implications of Heterogeneous Architectures

Figure 5a demonstrates that transitioning from a homogeneous mixture to a discrete layered architecture introduces a distinct neutronic penalty. Geometric separation can trap neutrons within isolated water channels and structural separators without ^{6}Li present like in homogenous models to capture the lower energy neutrons (Equation 2). The absorption heatmap (Figure 9b) confirms that these discrete separator ($\sim$20% of absorption) and coolant layers ($\sim$6% of absorption) act as pronounced parasitic sinks, driving overall neutron population loss. Importantly note that parasitic absorption within the water coolant is particularly problematic from a safety perspective; neutron capture and cross-contamination can produce tritiated water (HTO), which is approximately $10,000$ times more radiotoxic than elemental tritium gas (US Department of Energy, 2015).

On the other hand, the concentrated moderating effect of the water layers (Figure 9a) can increase the ^{6}Li breeding reaction (Equation 2) cross-sections in adjacent regions. As demonstrated in Figures 5b, 6a, and 6b, this thermalisation is so effective that the second breeder layer can actually yield a higher tritium production rate than the first, even though the total available flux has significantly decreased at that depth (Figure 7b).

Across the linear geometric gradient (ζ), the global TBR remains roughly constant, though it begins to marginally decline at higher, rear-weighted ζ values (Figure 6a). This relative stability indicates the blanket is operating in a 'saturating condition'; as visually confirmed by the fluence heatmap (Figure 7b), the total radial depth is sufficient to completely consume and attenuate the incident neutron flux regardless of internal distribution.

Total neutron population is dependent on $(\mathrm{n}, 2\,\mathrm{n})$ multiplication. These spallation events require high-energy incident neutrons (Equation 4). Neutron scattering and moderation decreases its energy as it travels through the blanket, causing multiplication to almost entirely deplete after the first two layers (Figure 8b). In rear-weighted configurations, incident neutrons must pass through structural separators and coolant channels earlier in their transport path. As shown by the scattering density (Figure 9a), this early moderation rapidly degrades neutron kinetic energy, explaining why total multiplication steadily decreases as ζ increases (Figure 8b).

This reduction in high-energy spallation directly links to the varying total flux tallies. Forward-weighted configurations sustain a significantly higher total flux (Figure 7a) which is partially due to the increased neutron population. However, the modest relative increase in multiplication (roughly $\sim$16.7%) is insufficient to fully account for the massive $\sim$54.5% increase in total flux track length (rising from $\sim$110 to $\sim$170 n·cm). The other contribution originates from forward-weighted designs lacking thick early water or separator layers to scatter the flux. With fewer early interactions, high-energy neutrons simply travel further before being thermalised, further increasing total flux track length for $\zeta < 0$ designs.

However, this does not mean that rear-weighted configurations are directly worse. High-energy neutrons exhibit higher scattering but lower absorption cross-sections in the mostly iron separators (Brown et al., 2018). This caused rear-weighted sepaarators to safely scatter fast neutrons with fewer immediate parasitic losses (Figure 10). This complex cross-sectional balance explains why forward-weighted designs, despite their vastly superior neutron populations, fail to achieve strictly dominant global TBR.

These geometric shifts also dictate material degradation. In rear-weighted configurations, the internal structural separators are exposed to high-energy source neutrons much earlier, driving a stark increase in internal DPA. Conversely, the FW experiences higher DPA in forward-weighted configurations, as intense multiplication within the thick first breeder layer causes highly energetic neutrons to backscatter. While the absorption heatmap (Figure 9b) demonstrates that zirconium is neutronically superior due to its exceptionally low parasitic absorption, there is a trade-off in its structural threshold. At our calculated damage rates zircaloy-4 restricts the FW lifetime to roughly 8–10 FPYs (6.2–7.5 DPA/year, critical helium embrittlement at 61.99 DPA (Gilbert et al., 2012)), far below ITER's 30-year operational target (Aymar, Barabaschi, and Shimomura, 2002). By contrast, a EUROFER97 FW would have vastly superior structural longevity of 23–26 FPYs (2.2–2.5 DPA/year, 57.47 threshold DPA (Woods, 1966)), but in testing we found would cause a $\approx 9\%$ TBR decrease.

Ultimately, navigating this web of physical variations is intractable via manual parameter sweeps. Due to coupled scaling of the coolant and breeder layers, manual ζ sweeps failed to unlock significant performance gains and resulted in largely stagnant global TBR and heating. To break this stagnation and automatically explore the highly-dimensional, non-linear design space, the demand and value of a Machine Learning optimisation pipeline became apparent.

6.2. Efficacy of the Machine Learning Optimisation

Despite the highly non-linear physical interactions governing the blanket in Section 6.1, the SOBO-GPR optimiser effectively trivialised the design space into a true 'corner solution'. Across multiple dimensional tests and constraint boundaries, the algorithm consistently identified that optimal performance dictates maximising breeder layer thicknesses while strictly minimising coolant channels (Section 5.2). The uniformity of this conclusion across various tests, summarised in Figure 12, strongly validates the robustness of both the surrogate model and the result. The specific converged thicknesses and TBR values were ultimately dictated by our applied bounds. The useful primary takeaway is the consistent trend towards 'binding limits', which was not obvious on manual investigation.

Crucially, the 2D contour maps (Figure 13) translate these mathematical optima into interpretable physics. Rather than identifying an isolated point within a 'black box', the contours demonstrate a continuous gradient towards the boundary constraints. Furthermore, the projections confirm that dimensional changes in the 'front' plasma-facing layers heavily dictate the TBR, whereas the outermost layers are largely inconsequential (Figures 13b, 13d). This arises from the physical reality established by the fluence mapping (Figure 7b); once the neutron population is totally attenuated, subsequent material distributions inherently cease to influence the objective function. The competing factors in Section 6.1 aggregate into a net result of thicker breeders (where a front loaded design would be taken if constrained) being preferable for higher TBR and thicker water coolant as ultimately harmful to TBR.

However, the 8D constrained optimisation (Figure 12) underscores a critical limitation of blind ML application. The algorithm converged on a seemingly complex, 'degenerate' result: an exponentially decaying thickness profile. While a correct optimum for the surrogate model, which merely maps an 8-dimensional thickness input to a scalar TBR output, nuclear analysis reveals this configuration is functionally identical to the standard binding constraints. Because the total flux is fully exhausted within B2, the precise distribution of the remaining rear volume is physically irrelevant to tritium production (affirmed by Figures 13b, 13d). This highlights that while Machine Learning is a powerful tool for navigating intractable parameter spaces, deriving accurate, actionable conclusions absolutely requires fundamental neutronic domain knowledge and follow-up interpretability studies.

6.3. Limitations and Future Work

While the SOBO-GPR approach successfully navigated the discrete blanket design space, the scope of this study presents several clear avenues for future investigation. Firstly, the extensive simulation dataset generated by the optimiser could be leveraged to fit an explicit analytic model. Rather than relying solely on GP, extracting coefficients from this data could quantitatively define the relative importance and potential couplings of specific structural layers.

Additionally, the findings here are specifically bound to a $Li_{17}Pb_{83}$ breeder and pressurised water coolant architecture. Alternative materials will undoubtedly exhibit different neutronic optima; however, the developed ML codebase (Khan, 2026b) is inherently modular. Alternative materials / configurations can be evaluated as 'drag-and-drop' objective functions, presenting an efficient mechanism for future large-scale material studies.

The most prominent limitation of the current framework is its reliance on Single Objective BO. Real-world blanket design is fundamentally a multi-physics compromise. The configurations in Figure 12 are indicative of singular TBR focus. Such breeder thicknesses could be impractical to implement and backloaded coolants contradict the primary heat extraction goals. As echoed by concurrent optimiser studies in the literature (Humphrey, Brooks, Mungale, et al., 2026), the critical next step is transitioning the pipeline to Multi-Objective Optimisation (MOO) (Emmerich, Giannakoglou, and Naujoks, 2006). By simultaneously mapping competing metrics, such as optimising TBR while also maximising heat extraction and minimising thickness / cost, the algorithm could generate balanced Pareto fronts Knowles, 2006. This evolution would upgrade the pipeline from a time-saving investigate simulation tool into a practical utility in fusion design decisions.

Furthermore, the successful convergence of the 8-dimensional constrained study demonstrates potential scalability of the algorithm to larger parameter spaces. Because the current simulations had manageable runtimes on standard desktop computing hardware (Section 4.3.1), transitioning the pipeline to High-Performance Computing (HPC) clusters could support significantly larger parameter spaces if more trials are needed. Even in lieu of enhanced computational bandwidth, the dimensionality of the model could be expanded to optimise the highly complex, fully asymmetric 3D blanket geometries of real designs and relax the layered blanket approximations. Finally, this scalability could provide a pathway to integrate broader operational variables (such as core plasma performance), facilitating a holistic, plant-wide fusion reactor optimisation model.

7. CONCLUSION

This study demonstrated that a heterogeneous layered breeder blanket degrades TBR due to the introduction of parasitic structural and coolant sinks. To navigate optimising the layer thicknesses, the bespoke SOBO-GPR pipeline proved highly effective, identifying that optimal TBR demands a front-loaded (lower ζ), 'binding configuration' of maximum breeder and minimum coolant thicknesses. However, this purely neutronic optimum does incur physical trade-offs and clearly reveals the negligible impact of the outermost blanket layers on overall production. Ultimately, while this study serves as a foundational introduction, these findings underscore that TBR cannot be isolated from structural and thermal-hydraulic realities, creating clear scope for future work to integrate these necessary design features. Because this successful implementation was built upon a modular codebase, there is substantial scope to broaden this approach to rapidly evaluate alternative materials and blanket architectures. As geometric approximations are subsequently relaxed, integrating this machine learning pipeline with larger parameter scaling and MOO would create a productive tool to support increasingly intractable design problems. This could assist in expediting the systematic and comprehensive research required for holistically optimised components in next-generation fusion reactors.

REFERENCES

Aymar, R., P. Barabaschi and Y. Shimomura (2002). 'The ITER design', *Plasma Physics and Controlled Fusion*, 44(5), pp. 519–565. DOI: 10.1088/0741-3335/44/5/304.

Balandat, M. et al. (2019). 'BoTorch: A Framework for Efficient Monte-Carlo Bayesian Optimization', *Advances in Neural Information Processing Systems 33, 2020*. DOI: 10.48550/arxiv.1910.06403.

Boccaccini, L. V. et al. (2014). 'Materials and design of the European DEMO blankets', *Journal of Nuclear Materials*, 455(1-3), pp. 136–140. DOI: 10.1016/j.jnucmat.2014.05.048.

Brown, D. et al. (2018). 'ENDF/B-VIII.0: The 8 th Major Release of the Nuclear Reaction Data Library with CIELO-project Cross Sections, New Standards and Thermal Scattering Data', *Nuclear Data Sheets*, 148, pp. 1–142. DOI: 10.1016/j.nds.2018.02.001.

Del Nevo, A. et al. (2019). 'Recent progress in developing a feasible and integrated conceptual design of the WCLL breeding blanket in Europe', *Fusion Engineering and Design*, 146, pp. 1805–1809. DOI: 10.1016/j.fusengdes.2019.03.040.

Emmerich, M. T., K. C. Giannakoglou and B. Naujoks (2006). 'Single-and multi-objective evolutionary optimization assisted by Gaussian random field metamodels'. *IEEE Transactions on Evolutionary Computation*. Vol. 10. 4. IEEE, pp. 421–439. DOI: 10.1109/TEVC.2005.859463.

Fausser, C., A. Li Puma and F. Gabriel (2012). 'Tokamak D-T neutron source models for different plasma physics confinement modes', *Fusion Engineering and Design*, 87(5-6), pp. 701–705. DOI: 10.1016/j.fusengdes.2012.02.025.

Federici, G. et al. (2017). 'European DEMO design strategy and consequences for materials', *Nuclear Fusion*, 57(9), pp. 1–26. DOI: 10.1088/1741-4326/57/9/092002.

Federici, G. et al. (2019). 'An overview of the EU breeding blanket design strategy as an integral part of the DEMO design effort', *Fusion Engineering and Design*, 141, pp. 30–42. DOI: 10.1016/j.fusengdes.2019.01.141.

Gilbert, M. R. et al. (2012). 'An integrated model for materials in a fusion power plant: transmutation, gas production, and helium embrittlement under neutron irradiation', *Nuclear Fusion*, 52(8), p. 083019. DOI: 10.1088/0029-5515/52/8/083019.

Gramacy, R. B. and H. K. Lee (2012). 'Cases for the nugget in modeling computer experiments', *Statistics and Computing*, 22(3), pp. 713–722. DOI: 10.1007/s11222-011-9251-5.

Heckrotte, W. and J. Hiskes (1971). 'Some factors in the choice of D-D, D-T or D-3He mirror fusion power systems', *Nuclear Fusion*, 11(5), pp. 471–484. DOI: 10.1088/0029-5515/11/5/009.

Hernández, F. and P. Pereslavtsev (2018). 'First principles review of options for tritium breeder and neutron multiplier materials for breeding blankets in fusion reactors', *Fusion Engineering and Design*, 137, pp. 243–256. DOI: 10.1016/j.fusengdes.2018.09.014.

Humphrey, L. et al. (2023). 'Machine learning techniques for sequential learning engineering design optimisation', *Plasma Physics and Controlled Fusion*, 66. DOI: 10.1088/1361-6587/ad0b05.

Humphrey, L., H. Brooks, S. Mungale, et al. (2026). 'Accelerating the engineering design of breeder blankets with parametric optimisation and sequential learning', *Frontiers in Nuclear Engineering*. DOI: 10.3389/fnuen.2025.1694684.

Kalos, M. H. and P. A. Whitlock (2008). *Monte Carlo Methods*. 1st ed. Wiley. DOI: 10.1002/9783527626212.

Khan, V. (2026a). 'Neutron Transport and Tritium Breeding Modelling in Nuclear Fusion Reactor Breeder Blankets'. MPhys Project Report. University of Manchester.

— (2026b). *VadanKhan/fusion-neutronics-modelling: MPhys Thesis Final Submission*. Version v1.0.0. DOI: 10.5281/zenodo.14867425.

Kirillov, I. et al. (1998). 'Liquid lithium self-cooled breeding blanket design for ITER', *Fusion Engineering and Design*, 39, pp. 863–869.

Knowles, J. (2006). 'ParEGO: A hybrid algorithm with on-line landscape approximation for expensive multiobjective optimization problems', *IEEE Transactions on Evolutionary Computation*, 10(1), pp. 50–66. DOI: 10.1109/TEVC.2005.851274.

Kovari, M. et al. (2018). 'Tritium resources available for fusion reactors', *Nuclear Fusion*, 58(2), pp. 1–10. DOI: 10.1088/1741-4326/aa9d25.

Kuridan, R. M. (2023). *Neutron Transport: Theory, Modeling, and Computations*. Graduate Texts in Physics. Cham: Springer Nature Switzerland. DOI: 10.1007/978-3-031-26932-5.

Lee, D. W. et al. (2012). 'Design of the HCCB TBM for ITER', *Fusion Engineering and Design*, 87(7-8), pp. 1167–1171.

Lord, M. et al. (2024). 'Fusing together an outline design for sustained fuelling and tritium self-sufficiency', *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 382(2280), pp. 1–15. DOI: 10.1098/rsta.2023.0410.

Mánek, P. et al. (2023). 'Fast regression of the tritium breeding ratio in fusion reactors', *Machine Learning: Science and Technology*, 4(1), p. 015008. DOI: 10.1088/2632-2153/acb6fb.

Maniscalco, J. A. et al. (1984). 'The Fusion Breeder — an Early Application of Nuclear Fusion', *Fusion Technology*, 6(3), pp. 584–596. DOI: 10.13182/FST84-A23140.

Moir, R. W. (1982). 'The fusion breeder', *Journal of Fusion Energy*, 2(4), pp. 351–367. DOI: 10.1007/BF01063686.

Nordlund, K. et al. (2018). 'Improving atomic displacement and replacement calculations with physically realistic damage models', *Nature Communications*, 9(1), p. 1084. DOI: 10.1038/s41467-018-03415-5.

Norgett, M. J., M. T. Robinson and I. M. Torrens (1975). 'A proposed method of calculating displacement dose rates', *Nuclear Engineering and Design*, 33(1), pp. 50–54. DOI: 10.1016/0029-5493(75)90035-7.

Olson, M. et al. (2025). 'Ax: A Platform for Adaptive Experimentation'. *AutoML 2025 ABCD Track*.

Paris, M. W. and M. B. Chadwick (2023). 'A lost detail in D–T fusion history', *Physics Today*, 76(10), pp. 10–11. DOI: 10.1063/PT.3.5317.

Paszke, A. et al. (2019). 'PyTorch: An Imperative Style, High-Performance Deep Learning Library', *arXiv*. DOI: 10.48550/arxiv.1912.01703.

Pearson, R. J., A. B. Antoniazzi and W. J. Nuttall (2018). 'Tritium supply and use: a key issue for the development of nuclear fusion energy', *Fusion Engineering and Design*, 136, pp. 1140–1148. DOI: 10.1016/j.fusengdes.2018.04.090.

Qiu, Y., L. Lu, L. V. Boccaccini, et al. (2018). 'Neutronic optimization and multi-physics analyses of the HCPB breeder blanket for EU DEMO', *Fusion Engineering and Design*, 136, pp. 920–925. DOI: 10.1016/j.fusengdes.2018.04.041.

Qu, S. et al. (2020). 'Neutronics effects of homogeneous model on solid breeder blanket of CFETR', *Fusion Engineering and Design*, 160, p. 111825. DOI: 10.1016/j.fusengdes.2020.111825.

Rasmussen, C. E. and C. K. Williams (2006). *Gaussian processes for machine learning*. Cambridge, MA: The MIT Press. DOI: 10.7551/mitpress/3206.001.0001.

Rastrigin, L. A. (1974). *Systems of extremal control*. Mir, Moscow.

Romano, P. K. et al. (2015). 'OpenMC: A state-of-the-art Monte Carlo code for research and development', *Annals of Nuclear Energy*, 82, pp. 90–97. DOI: 10.1016/j.anucene.2014.07.048.

Rosenbrock, H. H. (1960). 'An automatic method for finding the greatest or least value of a function', *The Computer Journal*, 3(3), pp. 175–184. DOI: 10.1093/comjnl/3.3.175.

Sawan, M. and M. Abdou (2006). 'Physics and technology conditions for attaining tritium self-sufficiency for the DT fuel cycle', *Fusion Engineering and Design*, 81(8), pp. 1131–1144. DOI: 10.1016/j.fusengdes.2005.07.035.

Schwefel, H.-P. (1981). *Numerical optimization of computer models*. John Wiley & Sons, Inc.

Segantin, S. et al. (2020). 'Optimization of tritium breeding ratio in ARC reactor', *Fusion Engineering and Design*, 154, pp. 1–5. DOI: 10.1016/j.fusengdes.2020.111531.

Shahriari, B. et al. (2016). 'Taking the human out of the loop: A review of Bayesian optimization', *Proceedings of the IEEE*, 104(1), pp. 148–175. DOI: 10.1109/JPROC.2015.2494218.

Shimwell, J. et al. (2021). 'The Paramak: automated parametric geometry construction for fusion reactor designs.', *F1000Research*, 10, p. 27. DOI: 10.12688/f1000research.28224.1.

Snoek, J., H. Larochelle and R. P. Adams (2012). 'Practical Bayesian optimization of machine learning algorithms'. *Advances in Neural Information Processing Systems*. Vol. 25. Curran Associates, Inc. DOI: 10.48550/arxiv.1206.2944.

Sorbom, B. et al. (2015). 'ARC: A compact, high-field, fusion nuclear science facility and demonstration power plant with demountable magnets', *Fusion Engineering and Design*, 100, pp. 378–405. DOI: 10.1016/j.fusengdes.2015.07.008.

Srinivas, N. et al. (2012). 'Information-theoretic regret bounds for Gaussian process optimization in the bandit setting', *IEEE Transactions on Information Theory*, 58(5), pp. 3250–3265.

Stornelli, G. et al. (2024). 'Ultra-fine grained EUROFER97 steel for nuclear fusion applications', *Journal of Materials Research and Technology*, 33, pp. 5075–5087. DOI: 10.1016/j.jmrt.2024.10.069.

Tanabe, T. (2017). *Tritium: Fuel of Fusion Reactors*. Tokyo: Springer. DOI: 10.1007/978-4-431-56460-7.

Trinkaus, H. (1985). 'Modeling of helium effects in metals: High temperature embrittlement', *Journal of Nuclear Materials*, 133-134, pp. 105–112. DOI: 10.1016/0022-3115(85)90119-9.

US Department of Energy (2015). *Tritium Handling and Safe Storage*. Tech. rep. DOE-STD-1129-2015. Washington, DC.

Warmer, F. et al. (2024). 'Overview of European efforts and advances in Stellarator power plant studies', *Fusion Engineering and Design*, 202, p. 114386. DOI: 10.1016/j.fusengdes.2024.114386.

Woods, C. R. (1966). *Properties of Zircaloy-4 tubing (LWBR Development Program)*. Tech. rep. WAPD-TM-585. Bettis Atomic Power Lab., Pittsburgh, PA (USA).

You, J.-H. et al. (2016). 'European baseline concept of the DEMO first wall: Design and manufacturing technologies', *Fusion Engineering and Design*, 109, pp. 1598–1603. DOI: 10.1016/j.fusengdes.2015.11.013.

Zhuang, G., G. Li, X. Jian, et al. (2019). 'Progress of the CFETR design', *Nuclear Fusion*, 59(11), p. 112010. DOI: 10.1088/1741-4326/ab0e27.

A. SUPPLEMENTARY THEORY

A.1. Nuclear Displacements and Damage Energy

In the standard Norgett-Robinson-Torrens (NRT) displacement model (Norgett, Robinson, and Torrens, 1975), the raw number of displaced atoms (N_d) is directly proportional to this damage energy:

$$N_{d,\mathrm{NRT}} = \frac{0.8 T_d}{2 E_d} \tag{10}$$

While the NRT model provides a reliable baseline, it often overestimates the final number of stable defects because it does not account for the athermal recombination of atoms that quickly regain their original lattice positions. To correct for this phenomenon, the Athermic Recombination-Corrected (ARC-DPA) model is employed (Nordlund et al., 2018). This introduces a surviving defect fraction, $\xi(T_d)$:

$$N_{d,\mathrm{arc}} = \frac{0.8 T_d}{2 E_d} \cdot \xi(T_d) = \frac{0.8 T_d}{2 E_d} \left((1-c) \left(\frac{0.8 T_d}{2 E_d} \right)^b + c \right) \tag{11}$$

where b and c are material-specific constants dependent on the displaced atom type. Values for the displacement energy (E_d) and these constants for critical structural materials, such as iron (dominating the structural separators) and zirconium (in the FW), are established in literature (Woods, 1966).

Finally, to determine if a material exceeds its irradiation limits over its operational lifetime, the total number of displacements must be normalised into DPA per Full Power Year (FPY). The DPA received by a structural component over a FPY is given by:

$$\mathrm{DPA/FPY} = \frac{N_{d,\mathrm{arc}}}{\left(\frac{\rho V \cdot N_A}{\mu} \right)} \cdot \frac{P}{E_f} \cdot (60^2 \cdot 24 \cdot 365) \tag{12}$$

Here, the denominator calculates the total number of target atoms within the specific component using the material density (ρ), volume (V), Avogadro's constant (N_A), and molar mass (μ). This is multiplied by the annual neutron emission rate, derived from the reactor thermal power (P) and the energy released per fusion event (E_f), scaled to the number of seconds in a calendar year.

A.2. Nuclear Heating and KERMA Factors

In neutronic simulations, localised energy deposition is calculated using KERMA (Kinetic Energy Released in MAterials) factors. The total nuclear heating rate, $H(\vec{r})$, at a specific spatial location is the integral of the local scalar flux $\Phi(\vec{r}, E)$ folded with the macroscopic Kerma factor $K(E)$ over all energies (Kuridan, 2023):

$$H(\vec{r}) = \int_E \Phi(\vec{r}, E) K(E) \, dE \tag{13}$$

where $K(E)$ accounts for the sum of all microscopic energy-releasing reaction cross-sections multiplied by the average energy released per reaction. Accurately mapping this heating distribution is critical, as it determines the required mass flow rates and spatial distribution of the coolant layers to prevent material melting and ensure efficient thermal energy extraction for the power cycle. Advanced Monte Carlo codes, such as OpenMC, directly implement these integrals via specialised heating tallies to provide high-fidelity energy deposition profiles (Romano et al., 2015).

A.3. Volume-Integrated Scalar Flux

The track-length estimator yields the volume-integrated scalar flux (Φ_V), which is expressed in units of neutron-centimetres (n · cm) (Kuridan, 2023; Kalos and Whitlock, 2008) and is formally defined as:

$$\Phi_V = \int_V \int_E \int_{4\pi} \psi(\vec{r}, \hat{\Omega}, E) \, d\hat{\Omega} \, dE \, dV = \sum_i W_i \ell_i \tag{14}$$

where W_i is the statistical weight of the neutron and ℓ_i is the path length of its trajectory through the volume V.

A longer total neutron path length implies a longer residence time within the breeder material, which directly increases the probability of nuclear interactions. The macroscopic reaction rate for any process, such as elastic scattering, is simply the product of the region's macroscopic cross-section (Σ_x) and this volume-integrated flux (Kuridan, 2023; Kalos and Whitlock, 2008).

A.4. Standard Acquisition Function: EI

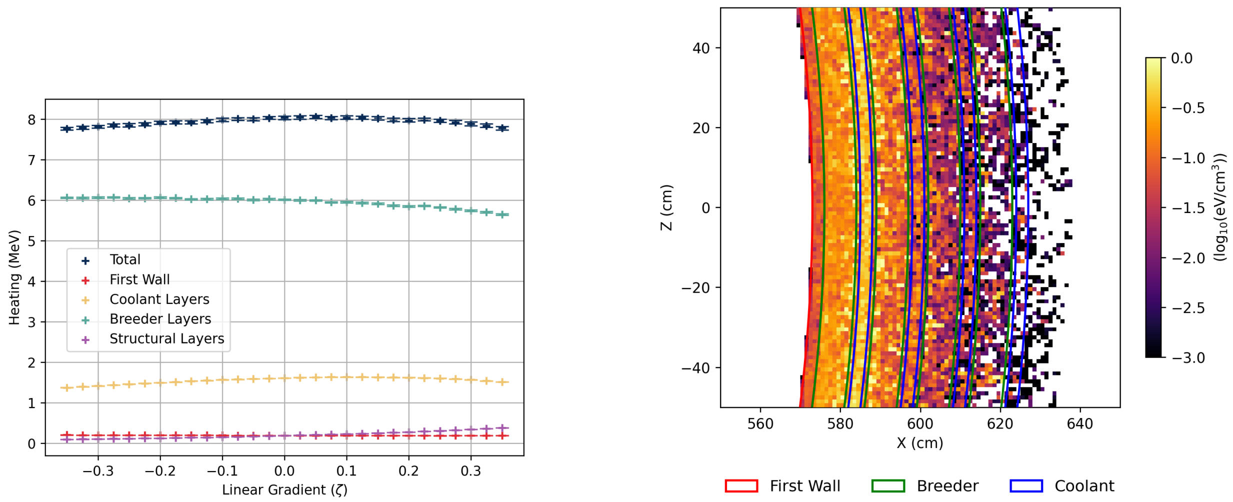
The most common standard acquisition metric is Expected Improvement (EI), which calculates the probability and magnitude by which a new sample might exceed the current best observation, f^*:

$$EI(x) = (\mu(x) - f^*)\Phi(Z) + \sigma(x)\phi(Z) \tag{15}$$

where Φ and ϕ are the standard normal cumulative distribution and probability density functions, respectively.

B. SUPPLEMENTARY NEUTRONICS DATA

This section provides the quantitative neutronic trends over the linear gradient parameter (ζ) and supplementary spatial heatmaps referenced in Section 5.1.



(a) Heating Energy vs ζ

(b) Heating Energy Density ($\zeta = 0$)

Figure B1: Total nuclear heating deposition rates and a corresponding spatial concentration within the discrete blanket layers.

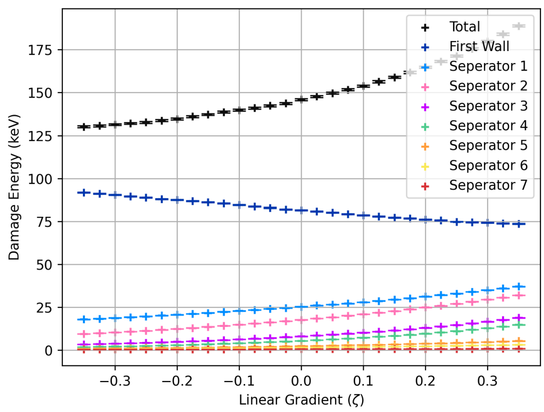


Figure B2: Total damage energy deposited into the plasma-facing FW and internal structural separators across the geometric gradient.